

8.2. Multiple Regression - Factorial Designs

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```
library(tidyverse)
library(car)
library(faraway)
library(GGally)
library(car)
library(multcomp)
```

Introduction

In this section we will illustrate how factorial designs can be analysed using the general linear model.

We will focus on a two-way anova design where we will model a continuous response with two factors.

Data

48 rats were allocated to

- 3 poisons (I,II,III) and
- 4 treatments (A,B,C,D),

and,

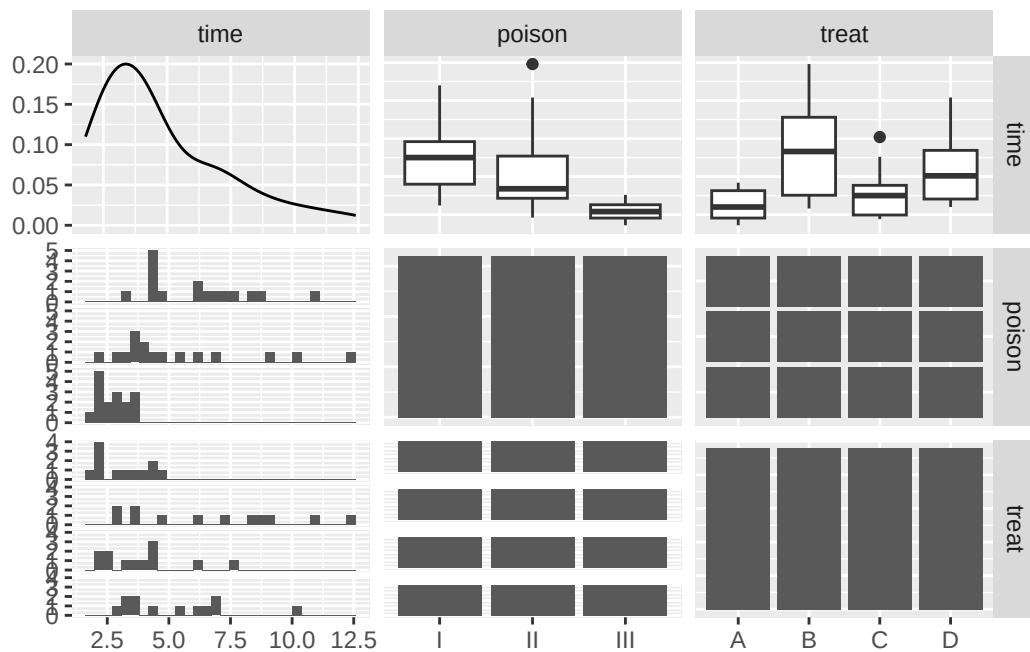
- the survival time was measured in (10 h)

We will first transform the data to hours.

```
data(rats)

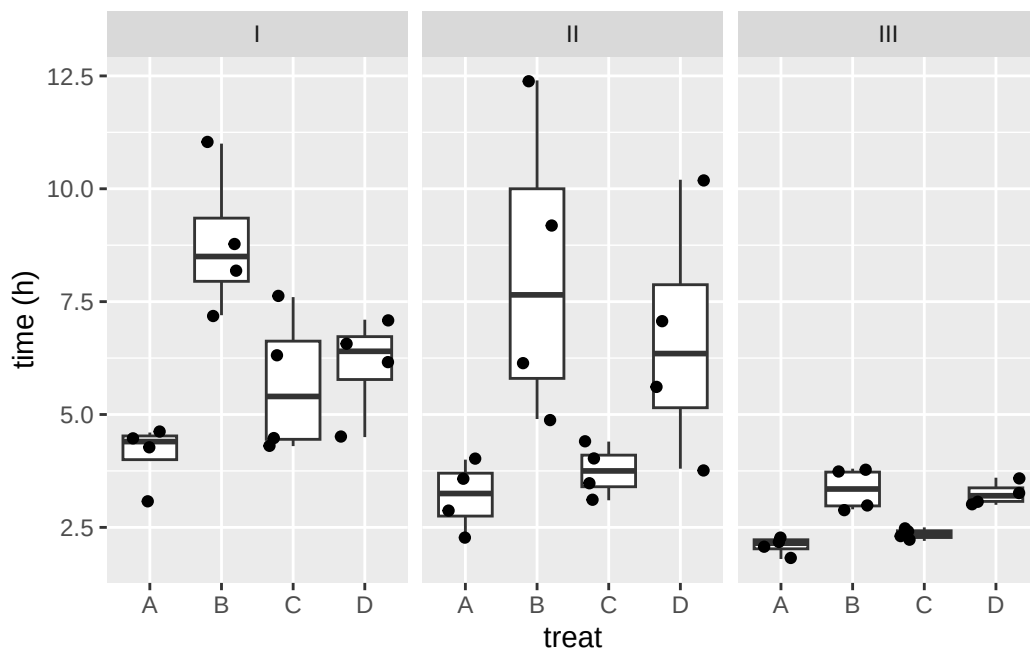
rats <- rats |>
  mutate(time = time * 10)

rats |>
  ggpairs()
```



The data exploration indicates that there seems to be an effect of both poison type and treatment.

```
rats |>
  ggplot(aes(x = treat, y = time)) +
    geom_boxplot(outlier.shape = NA) +
    geom_jitter() +
    facet_wrap(~poison) +
    ylab("time (h)")
```



- There might be an interaction, i.e. the effect of the treatment might be different according to the poison that has been adopted.
- The boxplots also indicate that the data are heteroscedastic.

Model

We will model the data with a main effect for poison and treatment and an poison $\times$ treatment interaction.

$$\begin{aligned}
 y_i = & \beta_0 + \beta_{II}x_{iII} + \beta_{III}x_{iIII} + \\
 & \beta_Bx_{iB} + \beta_Cx_{iC} + \beta_Dx_{iD} + \\
 & \beta_{II:B}x_{iII}x_{iB} + \beta_{II:C}x_{iII}x_{iC} + \beta_{II:D}x_{iII}x_{iD} + \\
 & \beta_{III:B}x_{iIII}x_{iB} + \beta_{III:C}x_{iIII}x_{iC} + \beta_{III:D}x_{iIII}x_{iD} + \epsilon_i
 \end{aligned}$$

with $i = 1, \dots, n$, $n = 48$, x_{iII} , x_{iIII} , x_{iB} , x_{iC} and x_{iD} dummy variables for poison II, III, treatment B, C, and D, respectively.

```
rats1 <- lm(time ~ poison * treat, rats)
summary(rats1)
```

Call:

```
lm(formula = time ~ poison * treat, data = rats)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.2500	-0.4875	0.0500	0.4312	4.2500

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	4.1250	0.7457	5.532	2.94e-06	***
poisonII	-0.9250	1.0546	-0.877	0.3862	
poisonIII	-2.0250	1.0546	-1.920	0.0628	.
treatB	4.6750	1.0546	4.433	8.37e-05	***
treatC	1.5500	1.0546	1.470	0.1503	
treatD	1.9750	1.0546	1.873	0.0692	.
poisonII:treatB	0.2750	1.4914	0.184	0.8547	
poisonIII:treatB	-3.4250	1.4914	-2.297	0.0276	*
poisonII:treatC	-1.0000	1.4914	-0.671	0.5068	
poisonIII:treatC	-1.3000	1.4914	-0.872	0.3892	
poisonII:treatD	1.5000	1.4914	1.006	0.3212	
poisonIII:treatD	-0.8250	1.4914	-0.553	0.5836	

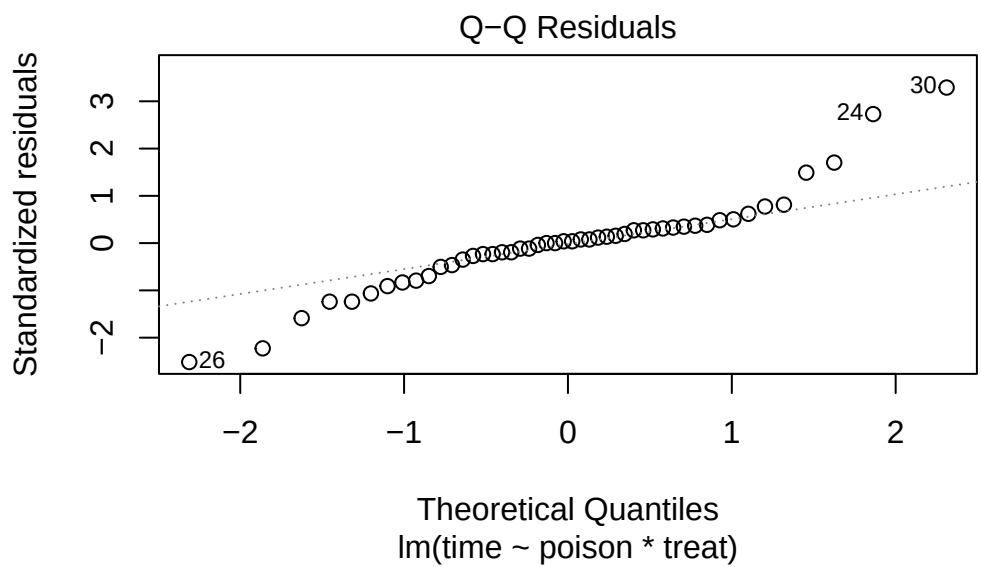
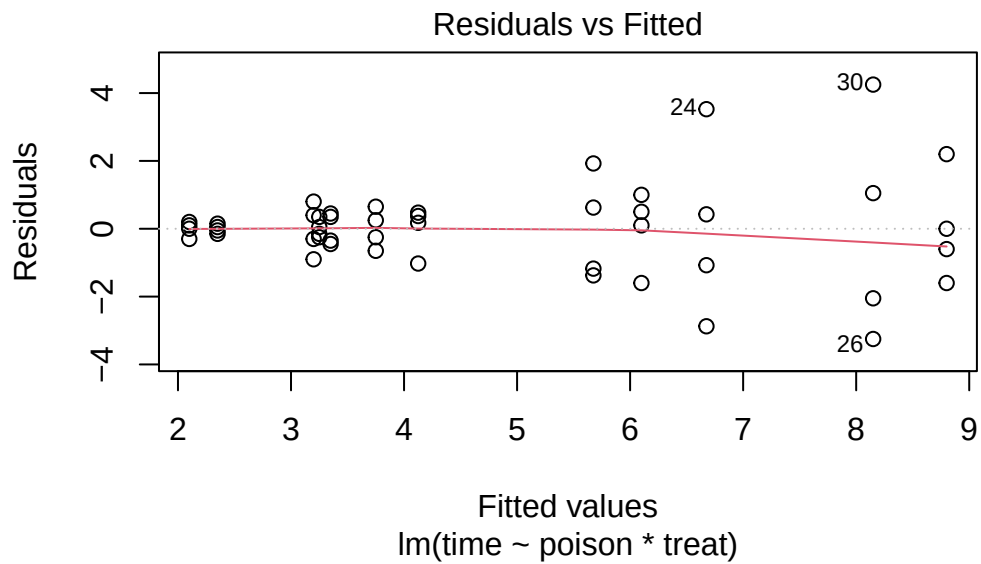
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

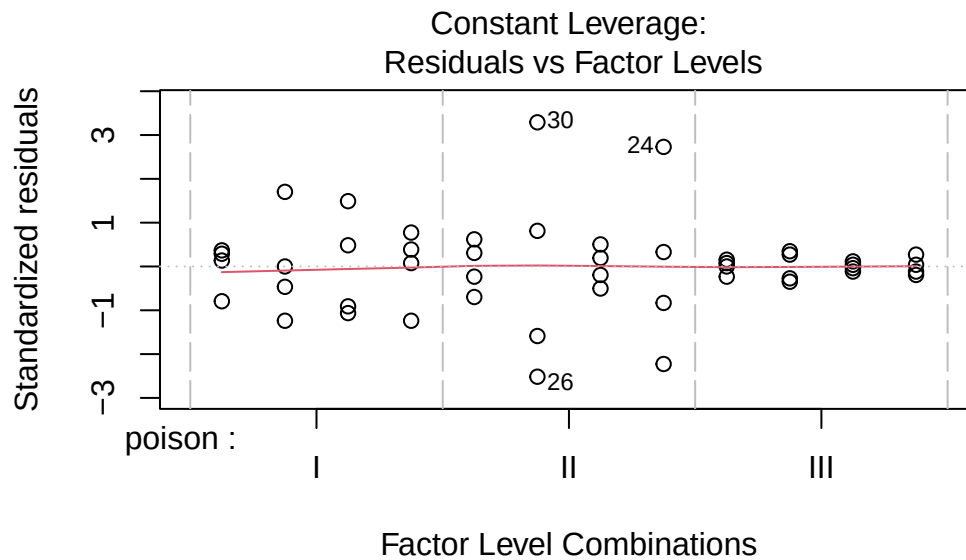
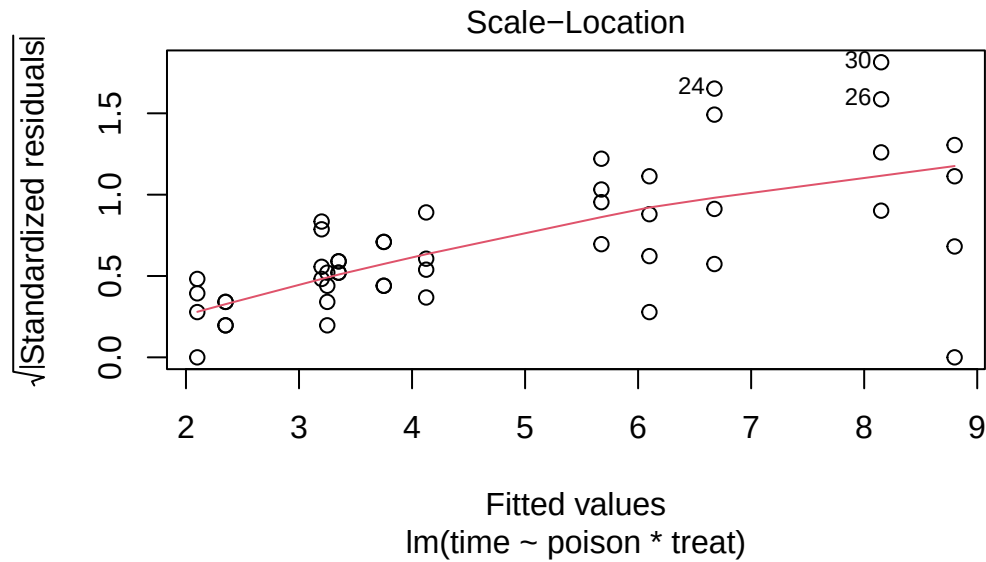
Residual standard error: 1.491 on 36 degrees of freedom

Multiple R-squared: 0.7335, Adjusted R-squared: 0.6521

F-statistic: 9.01 on 11 and 36 DF, p-value: 1.986e-07

```
plot(rats1)
```



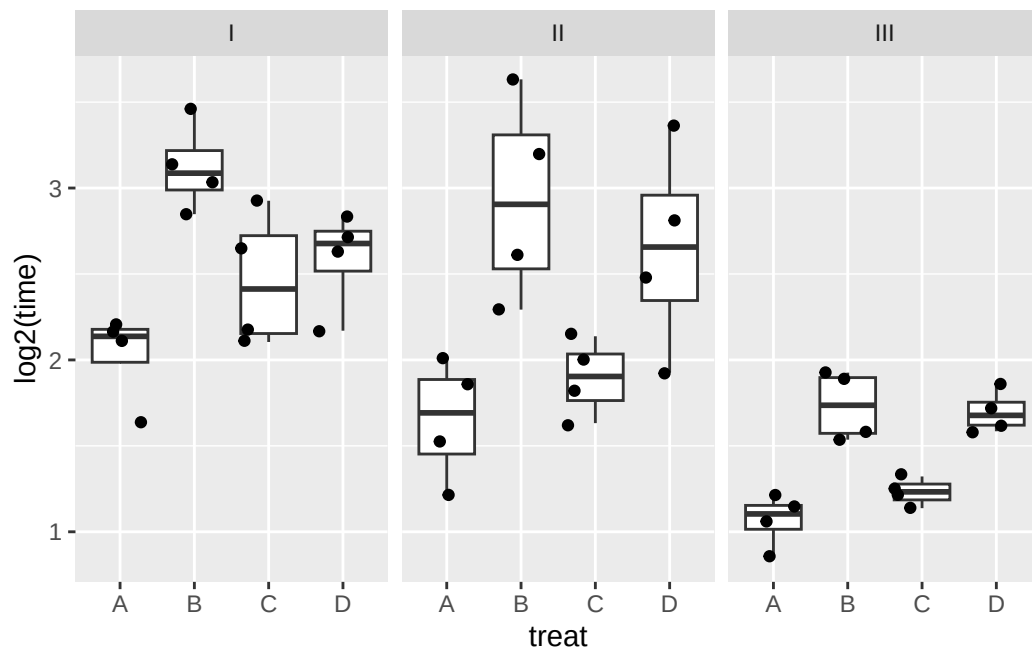


The errors, however, seem to be heteroscedastic and there seems to be a mean - variance relationship and they also appear to be distributed with broader tails than the normal distribution.

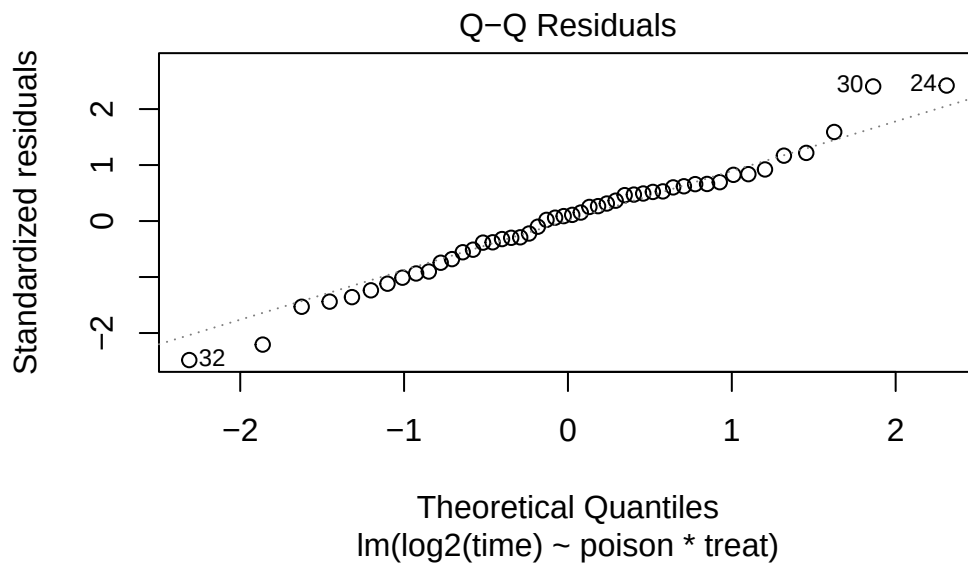
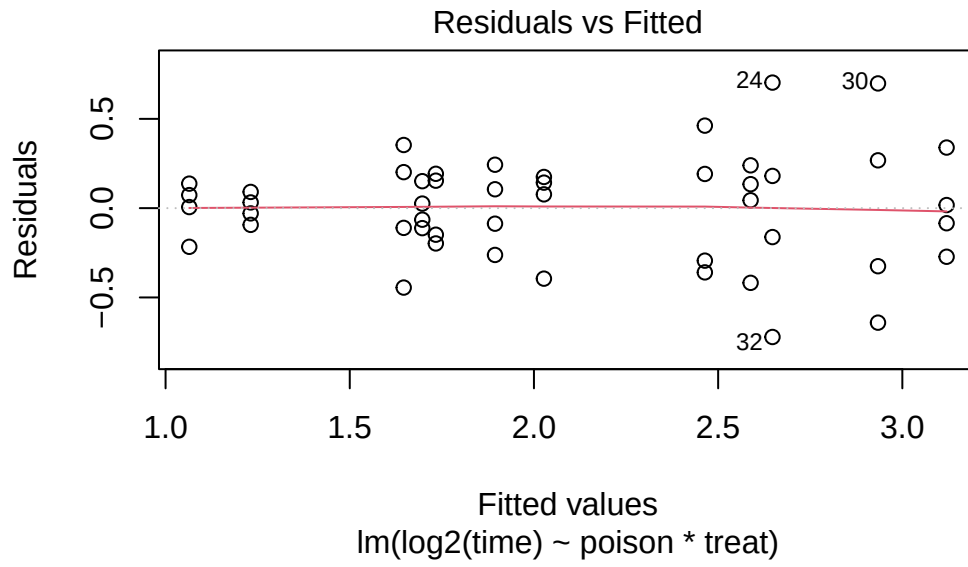
Transformations

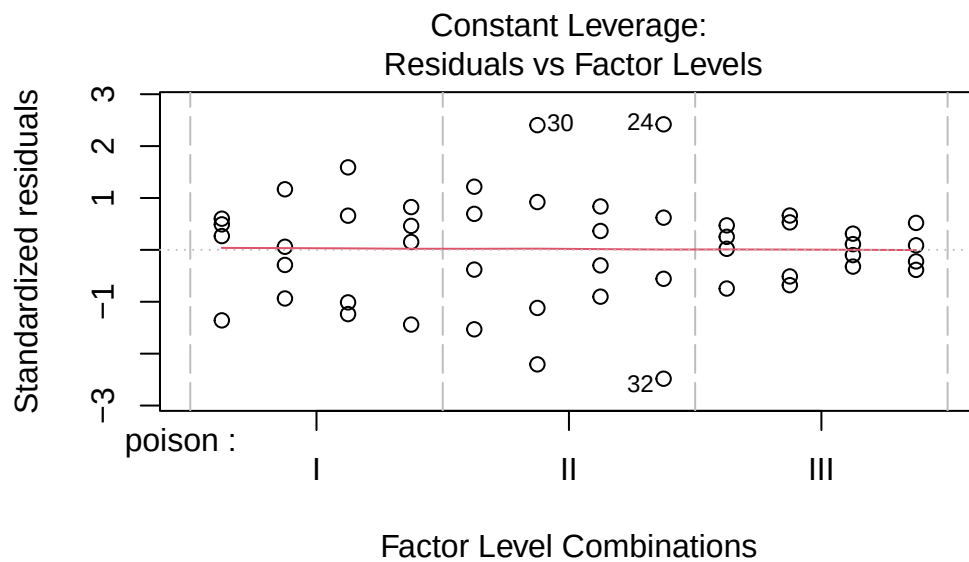
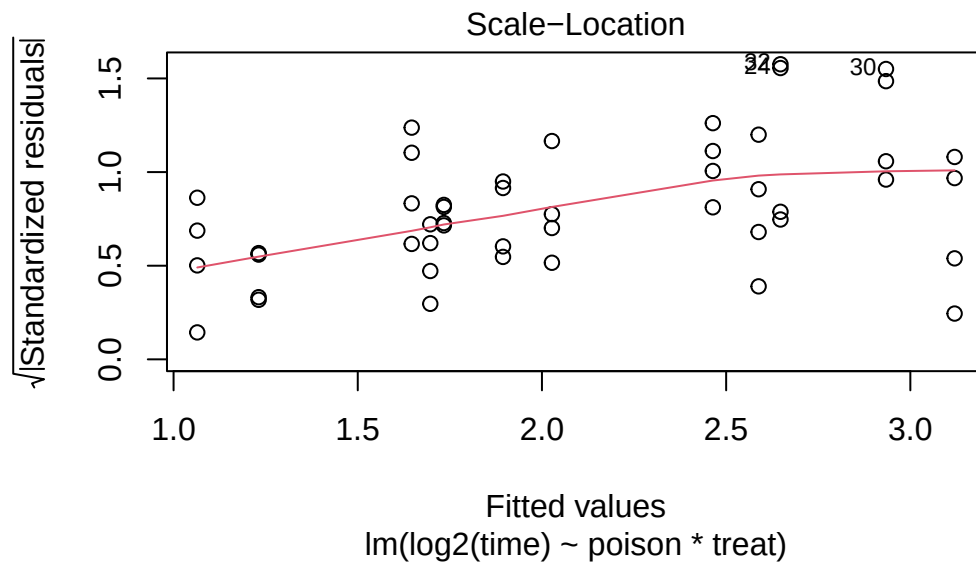
log transformation

```
rats |>  
  ggplot(aes(x = treat, y = log2(time))) +  
  geom_boxplot(outlier.shape = NA) +  
  geom_jitter() +  
  facet_wrap(~poison)
```



```
rats2 <- lm(time |> log2() ~ poison * treat, rats)  
plot(rats2)
```

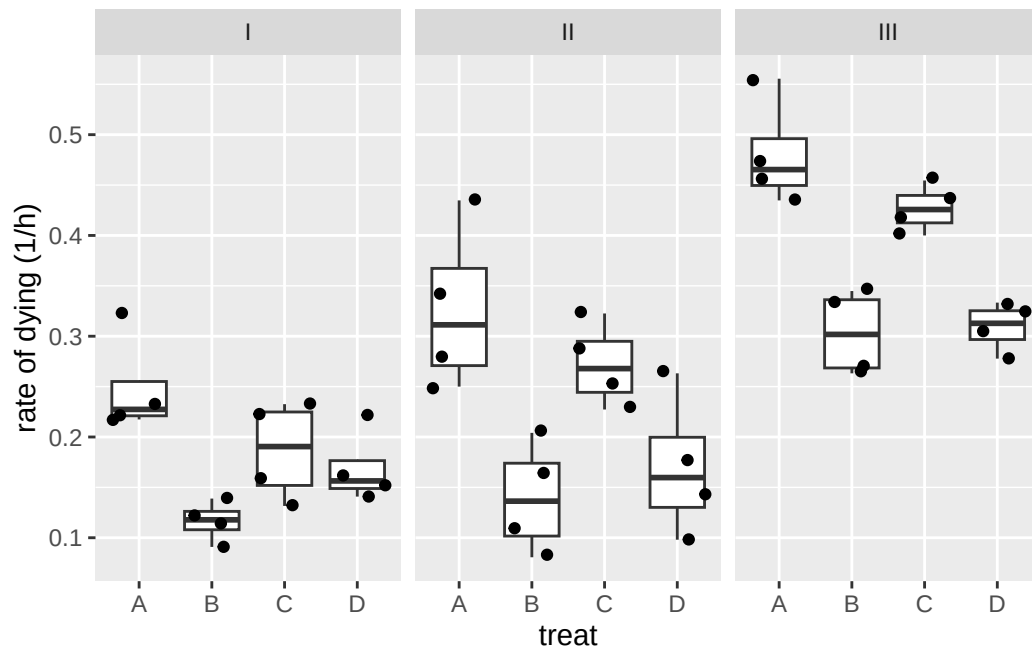




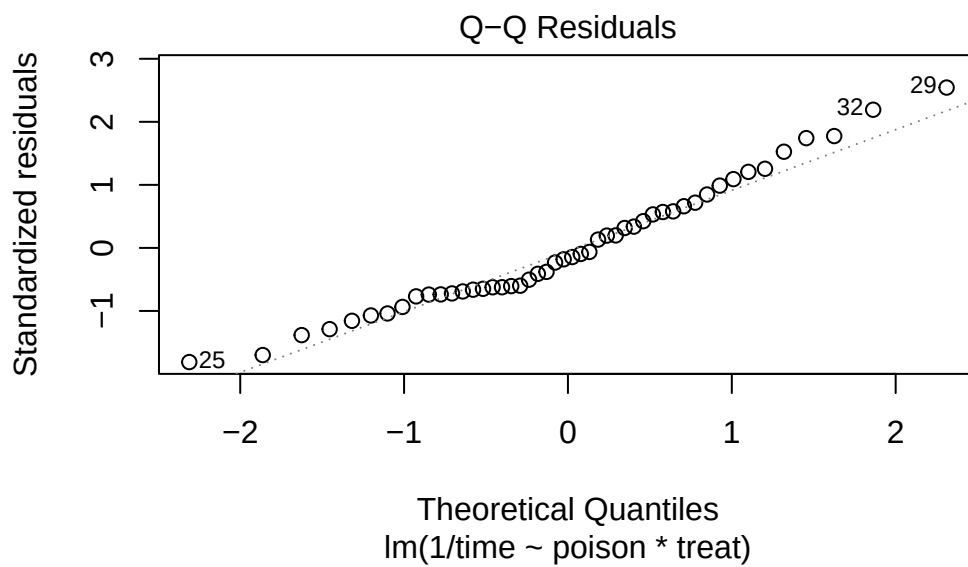
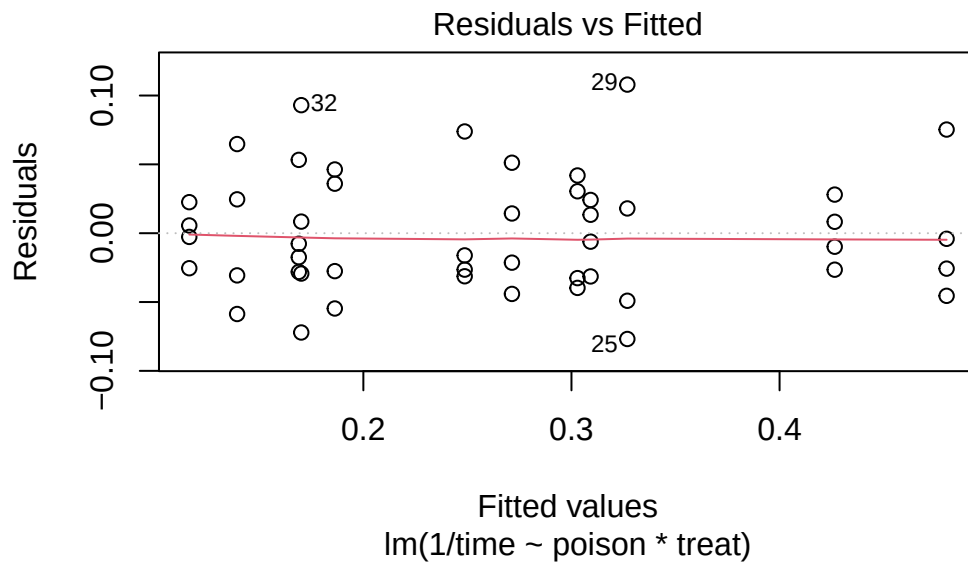
Log transformation does not remove the heteroscedasticity completely.

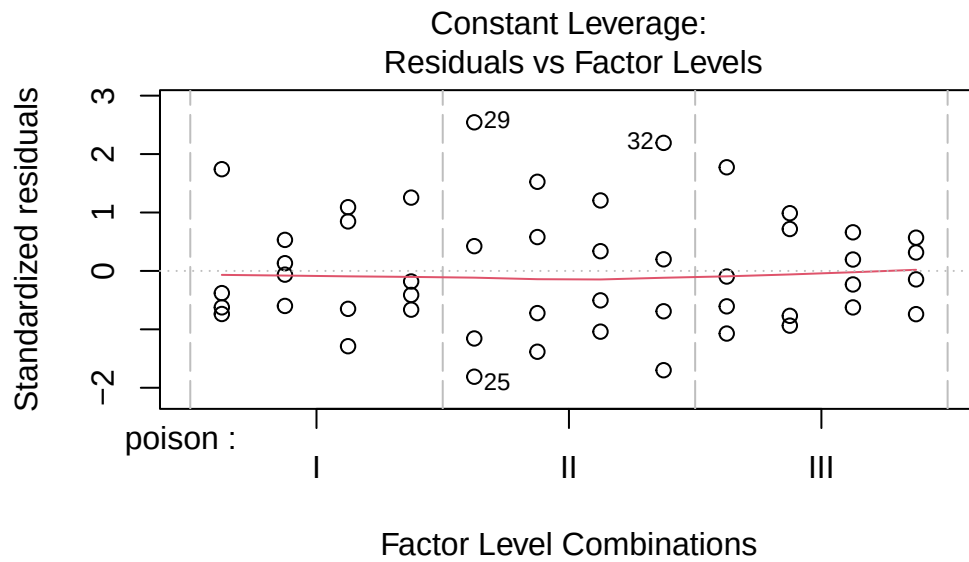
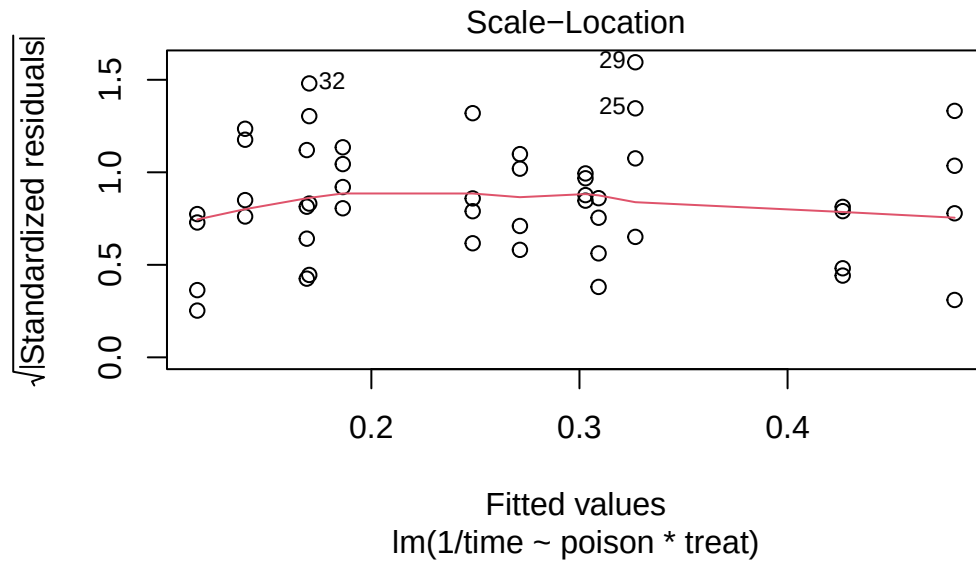
Reciprocal transformation

```
rats |>
  ggplot(aes(x = treat, y = 1 / time)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter() +
  facet_wrap(~poison) +
  ylab("rate of dying (1/h)")
```



```
rats3 <- lm(1 / time ~ poison * treat, rats)
plot(rats3)
```





The reciprocal transformation seems to do perform better and can be interpreted as the rate of dying.

Inference

There are multiple interaction terms involved in the factorial design. We will first assess them together, which can be done using the anova table.

```
Anova(rats3, type = "III")
```

Anova Table (Type III tests)

Response: 1/time

	Sum Sq	Df	F value	Pr(>F)	
(Intercept)	0.247383	1	103.0395	4.158e-12	***
poison	0.111035	2	23.1241	3.477e-07	***
treat	0.035723	3	4.9598	0.005535	**
poison:treat	0.015708	6	1.0904	0.386733	
Residuals	0.086431	36			

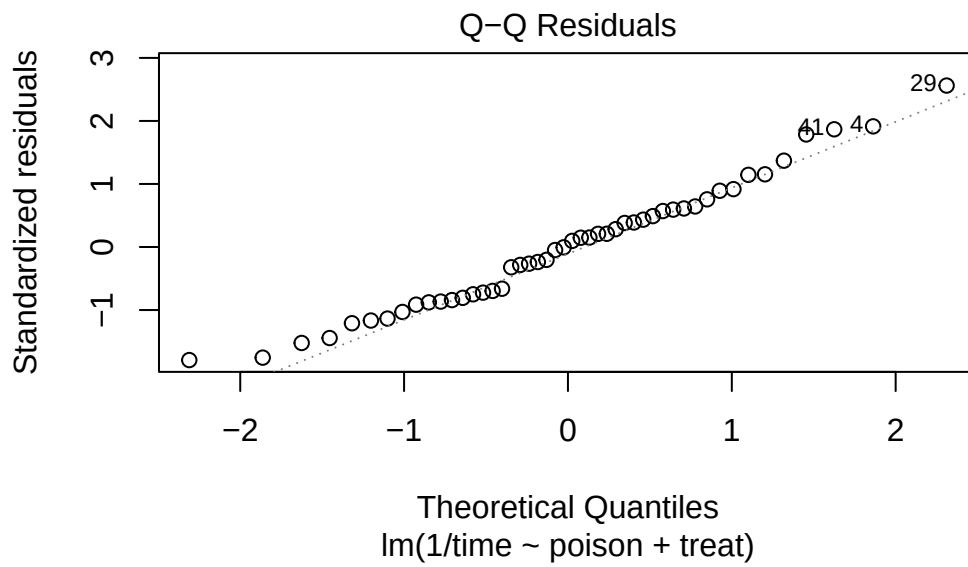
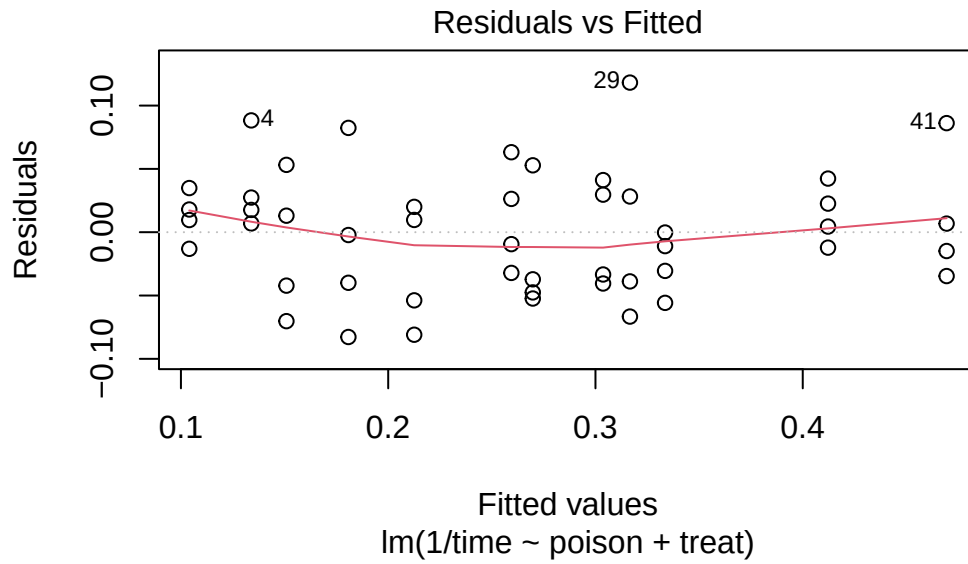
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

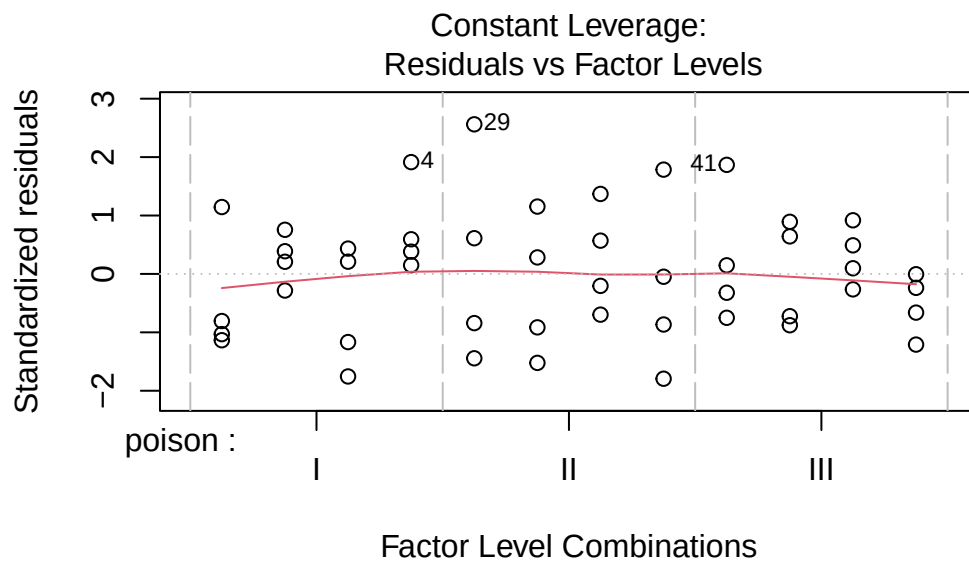
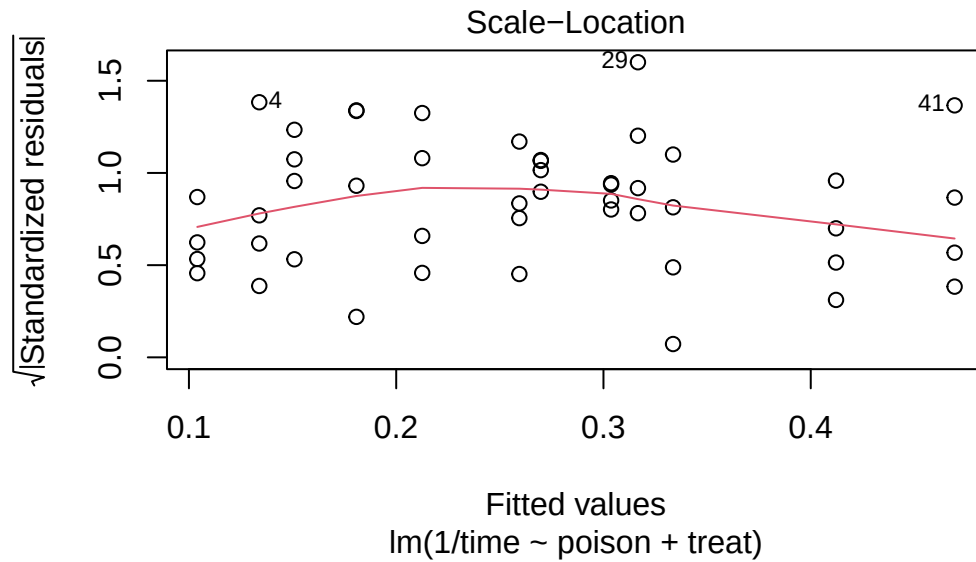
Removing the non-significant interaction term

The interaction appears to be not significant at the 5% level.

A common practice is to remove the interaction from the analysis. We then obtain an additive model and the effects of the two factors poison and treatment can be assessed separately.

```
rats4 <- lm(1 / time ~ poison + treat, rats)
plot(rats4)
```





```
Anova(rats4, type = "III")
```

Anova Table (Type III tests)

Response: 1/time

	Sum Sq	Df	F value	Pr(>F)
(Intercept)	0.58219	1	239.399	< 2.2e-16 ***
poison	0.34877	2	71.708	2.865e-14 ***
treat	0.20414	3	27.982	4.192e-10 ***
Residuals	0.10214	42		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The anova table shows that the effect of the poison and the treatment are both extremely significant ($p < 0.001$).

In the additive model we can assess the effect of the poison type and the treatments, separately in a post-hoc analysis.

```
comparisons <- glht(rats4, linfct = mcp(poison = "Tukey", treat = "Tukey"))
summary(comparisons)
```

Simultaneous Tests for General Linear Hypotheses

Multiple Comparisons of Means: Tukey Contrasts

Fit: `lm(formula = 1/time ~ poison + treat, data = rats)`

Linear Hypotheses:

	Estimate	Std. Error	t value	Pr(> t)
poison: II - I == 0	0.04686	0.01744	2.688	0.07344 .
poison: III - I == 0	0.19964	0.01744	11.451	< 0.001 ***
poison: III - II == 0	0.15278	0.01744	8.763	< 0.001 ***
treat: B - A == 0	-0.16574	0.02013	-8.233	< 0.001 ***
treat: C - A == 0	-0.05721	0.02013	-2.842	0.05080 .
treat: D - A == 0	-0.13583	0.02013	-6.747	< 0.001 ***
treat: C - B == 0	0.10853	0.02013	5.391	< 0.001 ***
treat: D - B == 0	0.02991	0.02013	1.485	0.61544
treat: D - C == 0	-0.07862	0.02013	-3.905	0.00282 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Adjusted p values reported -- single-step method)

```
confint(comparisons)
```

Simultaneous Confidence Intervals

Multiple Comparisons of Means: Tukey Contrasts

```
Fit: lm(formula = 1/time ~ poison + treat, data = rats)
```

Quantile = 2.8499

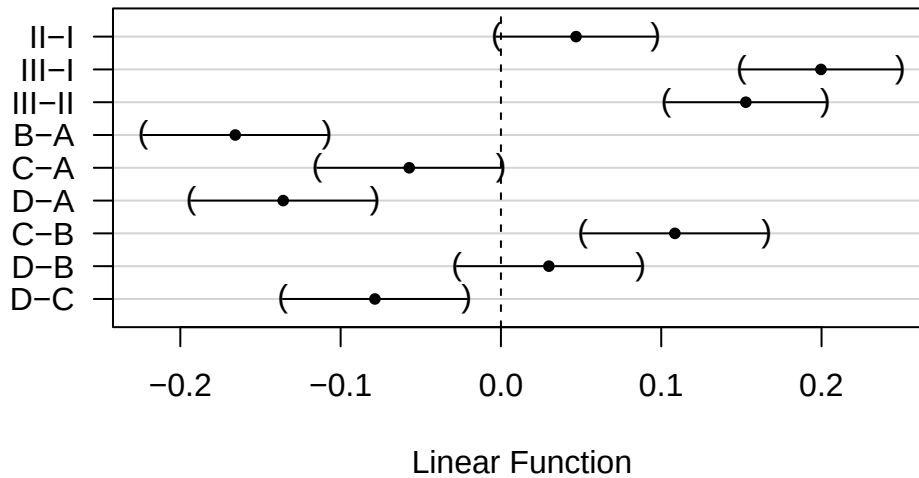
95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
poison: II - I == 0	0.0468641	-0.0028251	0.0965533
poison: III - I == 0	0.1996425	0.1499533	0.2493317
poison: III - II == 0	0.1527784	0.1030891	0.2024676
treat: B - A == 0	-0.1657402	-0.2231164	-0.1083641
treat: C - A == 0	-0.0572135	-0.1145897	0.0001626
treat: D - A == 0	-0.1358338	-0.1932100	-0.0784577
treat: C - B == 0	0.1085267	0.0511505	0.1659029
treat: D - B == 0	0.0299064	-0.0274698	0.0872826
treat: D - C == 0	-0.0786203	-0.1359965	-0.0212441

```
plot(comparisons, yaxt = "none")
contrastNames <- c("II-I", "III-I", "III-II", "B-A", "C-A", "D-A", "C-B", "D-B", "D-C")
axis(2, at = c(length(contrastNames):1), labels = contrastNames, las = 2)
```

95% family-wise confidence level



Keeping the non-significant interaction term

We know that accepting the null hypothesis that there is no interaction is a weak conclusion. It is possible that the experiment was simply underpowered to pick up the interaction. We can choose to keep the interaction in the model.

If the interaction were to be significant, this would mean that the effect of the poison changes according to the treatment and vice versa. Then, we cannot study the poison effects and treatment effects separately.

```
ExploreModelMatrix::VisualizeDesign(rats, ~ poison * treat)$plot
```

```
[[1]]
```

poison	III	(Intercept) + poisonIII	(Intercept) + poisonIII + treatB + poisonIII:treatB	(Intercept) + poisonIII + treatC + poisonIII:treatC	(Intercept) + poisonIII + treatD + poisonIII:treatD
	II	(Intercept) + poisonII	(Intercept) + poisonII + treatB + poisonII:treatB	(Intercept) + poisonII + treatC + poisonII:treatC	(Intercept) + poisonII + treatD + poisonII:treatD
	I	(Intercept)	(Intercept) + treatB	(Intercept) + treatC	(Intercept) + treatD
		A	B	C	D
		treat			

Hence, in case of a significant interaction we should study the effect of the poison for each treatment separately:

1. For treatment A we would have to assess the following comparisons:

- II-I: $H_0 : \beta_{II} = 0$
- III-I: $H_0 : \beta_{III} = 0$
- III-II: $H_0 : \beta_{III} - \beta_{II} = 0$

2. For treatment B we would have to assess the following comparisons:

- II-I: $H_0 : \beta_{II} + \beta_{II:B} = 0$
- III-I: $H_0 : \beta_{III} + \beta_{III:B} = 0$
- III-II: $H_0 : \beta_{III} + \beta_{III:B} - \beta_{II} - \beta_{II:B} = 0$

3. For treatment C we would have to assess the following comparisons:

- II-I: $H_0 : \beta_{II} + \beta_{II:C} = 0$
- III-I: $H_0 : \beta_{III} + \beta_{III:C} = 0$
- III-II: $H_0 : \beta_{III} + \beta_{III:C} - \beta_{II} - \beta_{II:C} = 0$

4. For treatment D we would have to assess the following comparisons:

- II-I: $H_0 : \beta_{II} + \beta_{II:D} = 0$
- III-I: $H_0 : \beta_{III} + \beta_{III:D} = 0$
- III-II: $H_0 : \beta_{III} + \beta_{III:D} - \beta_{II} - \beta_{II:D} = 0$

The same holds for assessing the effect of the treatment, which should be done for each poison separately:

1. Poison I

- B-A: $H_0 : \beta_B = 0$
- C-A: $H_0 : \beta_C = 0$
- D-A: $H_0 : \beta_D = 0$
- C-B: $H_0 : \beta_C - \beta_B = 0$
- D-B: $H_0 : \beta_D - \beta_B = 0$
- D-C: $H_0 : \beta_D - \beta_C = 0$

2. Poison II

- B-A: $H_0 : \beta_B + \beta_{II:B} = 0$
- C-A: $H_0 : \beta_C + \beta_{II:C} = 0$
- D-A: $H_0 : \beta_D + \beta_{II:D} = 0$
- C-B: $H_0 : \beta_C + \beta_{II:C} - \beta_B - \beta_{II:B} = 0$
- D-B: $H_0 : \beta_D + \beta_{II:D} - \beta_B - \beta_{II:B} = 0$
- D-C: $H_0 : \beta_D + \beta_{II:D} - \beta_C - \beta_{II:C} = 0$

3. Poison III

- B-A: $H_0 : \beta_B + \beta_{III:B} = 0$
- C-A: $H_0 : \beta_C + \beta_{III:C} = 0$
- D-A: $H_0 : \beta_D + \beta_{III:D} = 0$
- C-B: $H_0 : \beta_C + \beta_{III:C} - \beta_B - \beta_{III:B} = 0$
- D-B: $H_0 : \beta_D + \beta_{III:D} - \beta_B - \beta_{III:B} = 0$
- D-C: $H_0 : \beta_D + \beta_{III:D} - \beta_C - \beta_{III:C} = 0$

```
comparisonsInt <- glht(rats3, linfct = c(
  "poisonII = 0",
  "poisonIII = 0",
  "poisonIII - poisonII = 0",
  "poisonII + poisonII:treatB = 0",
  "poisonIII + poisonIII:treatB = 0",
  "poisonIII + poisonIII:treatB - poisonII- poisonII:treatB = 0",
  "poisonII + poisonII:treatC = 0",
  "poisonIII + poisonIII:treatC = 0",
  "poisonIII + poisonIII:treatC - poisonII- poisonII:treatC = 0",
  "poisonII + poisonII:treatD = 0",
  "poisonIII + poisonIII:treatD = 0",
  "poisonIII + poisonIII:treatD - poisonII- poisonII:treatD = 0",
  "treatB = 0",
```

```

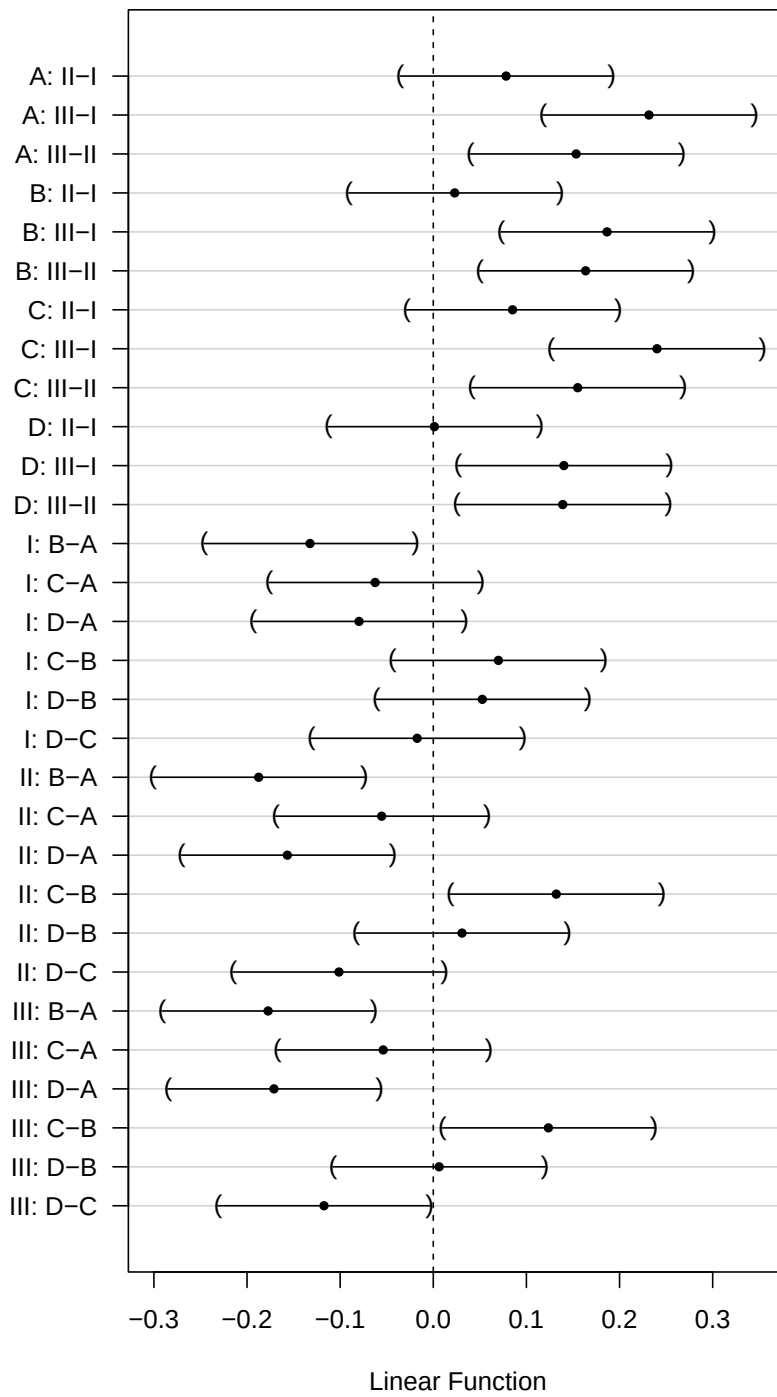
"treatC = 0",
"treatD = 0",
"treatC - treatB = 0",
"treatD - treatB = 0",
"treatD - treatC = 0",
"treatB + poisonII:treatB = 0",
"treatC + poisonII:treatC = 0",
"treatD + poisonII:treatD = 0",
"treatC + poisonII:treatC - treatB - poisonII:treatB = 0",
"treatD + poisonII:treatD - treatB - poisonII:treatB = 0",
"treatD + poisonII:treatD - treatC - poisonII:treatC = 0",
"treatB + poisonIII:treatB = 0",
"treatC + poisonIII:treatC = 0",
"treatD + poisonIII:treatD = 0",
"treatC + poisonIII:treatC - treatB - poisonIII:treatB = 0",
"treatD + poisonIII:treatD - treatB - poisonIII:treatB = 0",
"treatD + poisonIII:treatD - treatC - poisonIII:treatC = 0"
))

contrastNames <-
  c(
    paste(rep(LETTERS[1:4], each = 3), rep(apply(combn(c("I", "II", "III"), 2)[2:1, ], 2, pa
    paste(rep(c("I", "II", "III"), each = 6), rep(apply(combn(c(LETTERS[1:4])), 2)[2:1, ], 2,
  )

plot(comparisonsInt, yaxt = "none")
axis(2, at = c(length(contrastNames):1), labels = contrastNames, las = 2)

```

95% family-wise confidence level



Here, the interaction was not significant. Hence, the average effect of the poison type on the rate of dying does not change significantly according to the treatment and vice versa. In this case, it would make sense to estimate

1. the effect size for each pairwise comparison of poisons by averaging it over all treatments and
2. the effect size for each pairwise comparison of treatments by averaging it over all different poisons.

This should give us similar estimates as those obtained upon removing the interaction from the model.

e.g. for poison III vs poison II that would result in

- III-II:

$$H_0 : \frac{(\beta_{III} - \beta_{II}) + (\beta_{III} + \beta_{III:B} - \beta_{II} - \beta_{II:B}) + (\beta_{III} + \beta_{III:C} - \beta_{II} - \beta_{II:C}) + (\beta_{III} + \beta_{III:D} - \beta_{II} - \beta_{II:D})}{4}$$

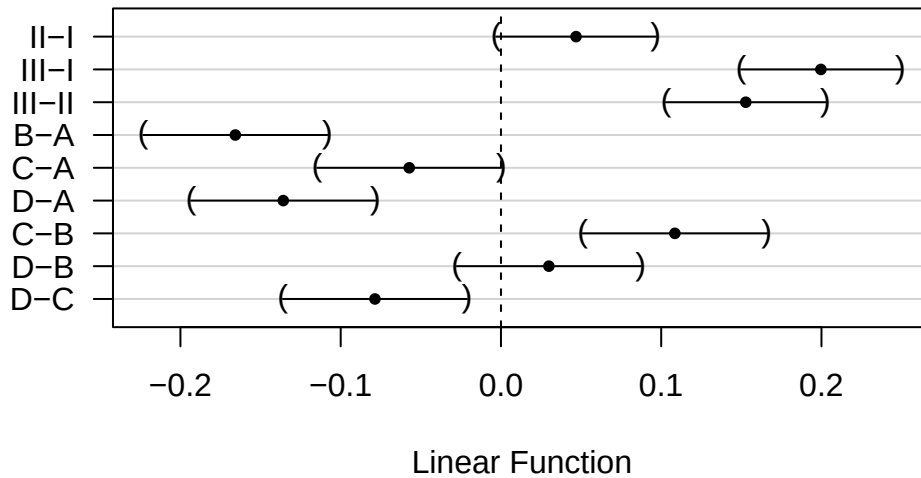
$$H_0 : \beta_{III} + \frac{1}{4} \times \beta_{III:B} + \frac{1}{4} \times \beta_{III:C} + \frac{1}{4} \times \beta_{III:D} - \beta_{II} - \frac{1}{4} \times \beta_{II:B} - \frac{1}{4} \times \beta_{II:C} - \frac{1}{4} \times \beta_{II:D} = 0$$

We can calculate the average contrast for each comparison of interest.

```
contrasts <- c(
  "poisonII + 1/4*poisonII:treatB + 1/4*poisonII:treatC + 1/4*poisonII:treatD = 0",
  "poisonIII + 1/4*poisonIII:treatB + 1/4*poisonIII:treatC + 1/4*poisonIII:treatD = 0",
  "poisonIII + 1/4*poisonIII:treatB + 1/4*poisonIII:treatC + 1/4*poisonIII:treatD - poisonII",
  "treatB + 1/3*poisonII:treatB + 1/3*poisonIII:treatB = 0",
  "treatC + 1/3*poisonII:treatC + 1/3*poisonIII:treatC = 0",
  "treatD + 1/3*poisonII:treatD + 1/3*poisonIII:treatD = 0",
  "treatC + 1/3*poisonII:treatC + 1/3*poisonIII:treatC - treatB - 1/3*poisonII:treatB - 1/3*",
  "treatD + 1/3*poisonII:treatD + 1/3*poisonIII:treatD - treatB - 1/3*poisonII:treatB - 1/3*",
  "treatD + 1/3*poisonII:treatD + 1/3*poisonIII:treatD - treatC - 1/3*poisonII:treatC - 1/3*"
)

comparisonsInt2 <- glht(rats3, linfct = contrasts)
plot(comparisonsInt2, yaxt = "none")
contrastNames <- c("II-I", "III-I", "III-II", "B-A", "C-A", "D-A", "C-B", "D-B", "D-C")
axis(2, at = c(length(contrastNames):1), labels = contrastNames, las = 2)
```

95% family-wise confidence level



Indeed, the effect sizes are exactly the same because the experiment is balanced.

Note, that the standard errors differ slightly. Indeed the errors of both models will differ as well as the remaining degrees of freedom of the errors $n - p$.

```
data.frame(Additive_coef = summary(comparisons)$test$coef, Additive_se = summary(comparisons)
```

	Additive_coef	Additive_se	Int_coef	int_se
poison: II - I	0.047	0.017	0.047	0.017
poison: III - I	0.200	0.017	0.200	0.017
poison: III - II	0.153	0.017	0.153	0.017
treat: B - A	-0.166	0.020	-0.166	0.020
treat: C - A	-0.057	0.020	-0.057	0.020
treat: D - A	-0.136	0.020	-0.136	0.020
treat: C - B	0.109	0.020	0.109	0.020
treat: D - B	0.030	0.020	0.030	0.020
treat: D - C	-0.079	0.020	-0.079	0.020

Conclusion

There is an extremely significant effect of the Poison and Treatment on the rate of dying of the rats ($p \ll 0.001$).

The effect of the poison does not significantly differ according to the treatment and vice versa ($p = 0.387$).

The rate of dying is on average 0.2h^{-1} and 0.15h^{-1} higher for rats that are exposed to poison III than those exposed to poison I and II, respectively (both $p \ll 0.001$, 95% CI III-I: $[0.15, 0.25]\text{h}^{-1}$, 95% CI III-II: $[0.1, 0.2]\text{h}^{-1}$). The average rate of dying was not significantly different between rats exposed to poison I and poison II ($p=0.074$).

The rate of dying is on average 0.17h^{-1} and 0.14h^{-1} higher upon treatment A than upon treatment B and D, respectively ($p \ll 0.001$, 95% CI B-A: $[-0.22, -0.11]\text{h}^{-1}$, 95% CI D-A: $[-0.19, -0.08]\text{h}^{-1}$). The rate of dying is on average 0.11h^{-1} and 0.08h^{-1} higher upon treatment C than upon treatment B and D, respectively (C-B: $p \ll 0.001$, 95% CI $[0.05, 0.17]\text{h}^{-1}$, D-C: $p = 0.003$, 95% CI $[-0.14, -0.02]\text{h}^{-1}$). The average rate of dying was not significantly different between treatment C and A ($p = 0.051$), and, between treatment B and D ($p = 0.61$).

All p-values are corrected for multiple testing.